# SNL DES SGU BPM II DM Practice Exam

*The following document contains questions from the DM module that were 1) originally created as well as 2) drawn from various sources such as textbooks, other DES group pages etc. It is 116 questions long and <u>is in our opinion</u> similar in style and content to the real exam. It is intended to be used as a practice tool. Do not just focus on these practice questions for the exam – ensure that all material is reviewed sufficiently.*

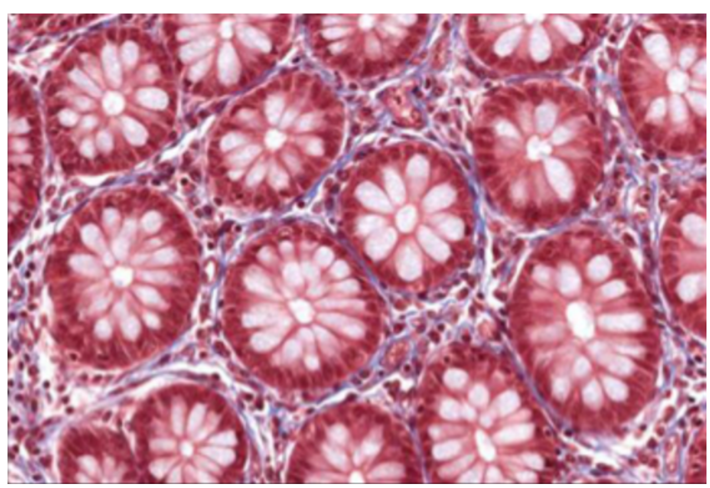
1. The death of Wolfgang Amadeus Mozart has always been a great mystery. Modern investigators have dug up the corpse and found large traces of arsenic. The aforementioned compound inhibits which of the following enzymes?
(A) Pyruvate Dehydrogenase
(B) Aconitase
(C) Glyceraldehyde 3-phosphate Dehydrogenase
**(D) A & C**
(E) All of the above

**Explanation**: Arsenic inhibits lipoic acid, a cofactor in dehydrogenase enzymes. Arsenic also inhibits Alpha-Ketoglutarate Dehydrogenase.

2. A 44-year old man presents with moon face, a buffalo hump, central obesity and hyperpigmentation of the skin. He is diagnosed with Cushing's disease. The hormone causing these symptoms induces which one of the following steps of gluconeogenesis?
(A) Pyruvate → Oxaloacetate
**(B) Oxaloacetate → Phosphoenolpyruvate**
(C) Fructose-1,6-bisphosphate → Fructose-6-phosphate
(D) Glucose-6-phosphate → Glucose

**Explanation**: Cushing's disease is a result of an ACTH-secreting pituitary tumor, which would result in elevated cortisol levels. Cortisol is an important inducer of PEP carboxykinase, which catalyzes the step Oxaloacetate → PEP

3. A 49-year old male patient comes to the physician complaining of pain in his lower left abdominal area. A biopsy is taken and the following histological slide is seen. Which of the following locations has these features?



A. Duodenum
B. Ileum
C. Jejunum
**D. Large intestine**
E. Appendix

4. A 67-year-old man with a history of alcohol abuse presents to the emergency room with severe epigastric pain, hypotension, abdominal distension, and diarrhea with steatorrhea. Serum amylase and lipase are found to be greater than normal, leading to a diagnosis of pancreatitis. The steatorrhea can be accounted for by a decrease in the intraluminal concentration of which of the following pancreatic enzymes?
A. Trypsin
B. Chymotrypsin
C. Colipase
D. Amylase
**E. Lipase**

**Explanation:** Steatorrhea = fatty stools, resulting from an inability to metabolize fat. This may be caused by a deficiency of lipase.

5. A 55-year old man was admitted to the hospital with severe abdominal pain. Gastroscopy and CT scan examinations revealed a perforating ulcer in the posterior wall of the stomach. Where would peritonitis most likely develop initially?
A. Greater sac
**B. Omental bursa**
C. Right subhepatic space
D. Right subphrenic space
E. Hepatorenal space

**Explanation:** The omental bursa is also known as the lesser sac, and is located in the posterior region.

6. A 27-year old woman goes on a five day fast, only intermittently sipping from a 6oz bottle of orange juice. Which of these enzymes would most likely be active during this period?
A. Glycogen Synthase
B. PFK-1
**C. Fructose-1,6-bisphosphatase**
D. Acetyl-CoA Carboxylase
E. Pyruvate Kinase

**Explanation:** Despite the intermittent sipping of a small amount of orange juice, the woman is still in the fasted state. As a result, fructose-1,6-bisphosphate, the regulated enzyme of gluconeogenesis, will be active.

7. A 22-year old woman presents with deep facial flushing due to accidental ingestion of a toxin. The toxin in question inhibits cytochrome oxidase of the electron transport chain. What is the likely identity of this toxin?
**A. Azide**
B. Rotenone
C. 2,4-Dinitrophenol
D. Oligomycin
E. Bongkrekic Acid

8. A physician is giving a seminar on alcohol abuse and the best ways to treat patients. He discusses at length the dangers of a drug that causes a buildup of alcohol byproducts that leads to extremely unpleasant adverse effects, including flushing, headache, diaphoresis, nausea, and vomiting. Which enzyme does this drug likely inhibit?
A. CYPE21
B. Alcohol dehydrogenase
**C. Acetaldehyde dehydrogenase**
D. Pyruvate dehydrogenase
E. Lactate dehydrogenase

**Explanation:** The drug in question is likely disulfiram, which inhibits acetaldehyde dehydrogenase.

9. A 58-year old man had been scheduled for a cholecystectomy. During the operation the scissors of the surgical resident accidentally entered the tissues immediately posterior to the epiploic (omental) foramen. The surgical field was filled immediately with profuse bleeding. Which of the following vessels was most likely the source of bleeding?
A. Portal vein
**B. Inferior vena cava**
C. Aorta
D. Right renal artery
E. Superior mesenteric vein

10. A 28-year old woman refuses to eat for 2 weeks in order to lose weight to fit into the dress she purchased for her upcoming wedding. Examination of her breath shows a fruity odor. Which one of these enzymes is the rate-determining enzyme of the primary source of energy for her brain at this time?
A. Glucose-6-phosphatase
**B. HMG-CoA synthase**
C. Fructose-1,6-bisphosphate
D. HMG-CoA reductase
E. Acetyl-CoA carboxylase

**Explanation:** After such an extended period of fasting, the primary source of energy for the woman's brain is through ketogenesis (hint in the stem was a fruity breath, due to acetone). The rate-determining enzyme of ketogenesis is HMG-CoA synthase.

11. A 38-year old woman comes to her physician complaining about "her age getting to her." She states that she feels tired more often and finds it difficult to concentrate and focus at work lately. She says that she read that antioxidant vitamins can help with these symptoms, but isn't sure which supplements she should be taking. What should the physician recommend as the best antioxidant vitamins?
A. Vitamins A, K and C
B. Vitamins K, D and E
C. Vitamin C and thiamine
**D. Vitamin E, A and C**

E. Vitamin E, C and thiamine

12. A child is brought to the pediatrician because her parents are concerned about lead poisoning since their house is known to contain leadbased paint. A complete blood count reveals anemia. Lead poisoning causes anemia because it does which of the following?
**A. Disrupts heme synthesis by inhibiting ferrochelatase**
B. Disrupts heme synthesis by increasing the activity of aminolevulinate dehydratase
C. Disrupts heme synthesis by causing decreased iron absorption from the gut
D. Disrupts hemoglobin function by binding to hemoglobin with high affinity, preventing oxygen binding
E. Disrupts RBC DNA synthesis, causing megaloblastic changes in RBCs

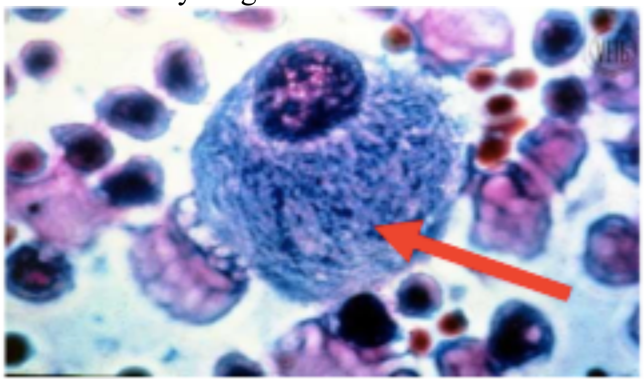
13. A 21-year old male is imaged following an injury while camping. A laceration of the body of the pancreas is seen with hemorrhage around the vessel traveling along the superior border of the pancreas. Which vessel is this?
A. Left gastric artery
B. Portal vein
**C. Splenic artery**
D. Left gastro-omental artery
E. Superior mesenteric artery

14. A 25-year old woman presents to the ER complaining of nausea, vomiting, and abdominal distension. Diagnostic tests reveal the image below. What region of the GI tract is most likely affected?


A. Esophagus
B. Stomach
**C. Small Intestine**
D. Large Intestine
E. Rectum

**Explanation:** “Stack of coins” image is due to small bowel obstruction.

15. A 13-year old girl presents to the physician with weak bones, hepatosplenomegaly and mild developmental delay. A histological image of her cells is prepared and displayed below. What is her most likely diagnosis?



A. Hurler's Syndrome
B. Von Gierke's Disease
C. Hunter Syndrome
**D. Gaucher's Disease**
E. Tay Sachs

**Explanation:** The image shows "crumpled tissue paper" morphology, which is characteristic of Gaucher's disease.

16. A woman and her husband both have a waist circumference of 38 inches. Compared to her husband, the woman would most likely have which of the following?
**A. A greater proportion of visceral fat**
B. Increased insulin sensitivity
C. A greater proportion of subcutaneous fat
D. Increased HDL levels
E. Decreased LDL levels

**Explanation:** Women with waist circumferences ≥ 35 inches are susceptible to metabolic syndrome while men with waist circumferences ≥ 40 inches are susceptible to metabolic syndrome. Individuals with metabolic syndrome generally have more visceral fat, lower HDL levels and decreased insulin sensitivity.

17. A 72-year old man living on his own is brought to the emergency department after his son found him confused and flustered in his apartment. The physician notes that there are plaques on his body and is severely dehydrated, likely caused by his diarrhea. Which of the following deficiencies does this patient likely have?
A. Folate
B. Iron
C. Cobalamin
**D. Niacin**
E. Riboflavin

**Explanation:** Niacin deficiency may lead to Pellagra, characterized by the 4 D's: Dermatitis, Dementia, Diarrhea and (potentially) Death.

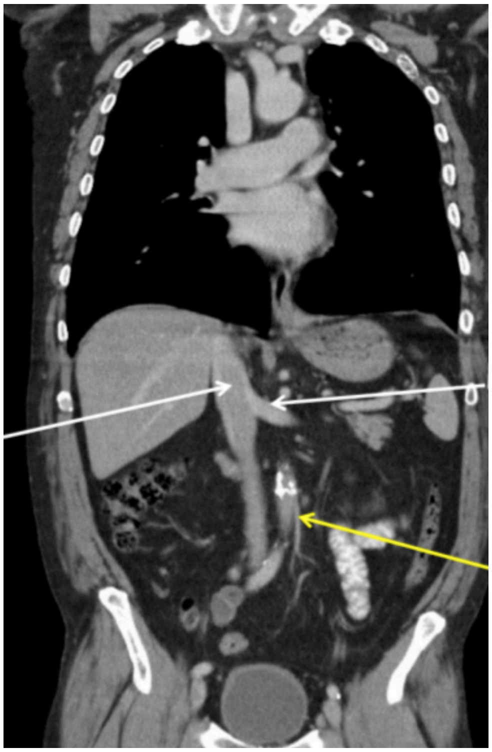
18. A 58-year old woman underwent a gastroscopy, which revealed a tumor in the pyloric antrum of the stomach. A CT scan was ordered to evaluate lymph node involvement. Which of the following nodes initially drains the pyloric antrum?
A. Hepatic
B. Lumbar
**C. Celiac**
D. Superior mesenteric
E. Inferior mesenteric

19. A 32-year old woman is brought to a Caribbean emergency department by her husband. She is unconscious and presents as severely hypoglycemic. The husband explains to the physician that they were on vacation and were trying different vegetables and fruits. Which of following will most likely be seen in a urine sample of the patient?
A. Increased long chain acyl carnitines
**B. Increased medium chain acyl carnitines**
C. Decreased medium chain acyl carnitines
D. Decreased dicarboxylic acids

**Explanation:** The patient likely ingested the unripe ackee fruit, which contains hypoglycin A, an inhibitor of MCAD. Medium chain acyl carnitines would therefore not be able to be metabolized and would appear in the urine.

20. An 83-year-old woman with constipation is prescribed a high-fiber diet, which leads to an increased production of short-chain fatty acids (SCFAs). SCFA absorption occurs almost exclusively from which of the following segments of the GI tract?
A. Stomach
B. Duodenum
C. Jejunum
D. Ileum
**E. Colon**

21. A 42-year presents with abdominal pain. An MRI is taken in the coronal view. The cause of the abdominal pain is not found, but the physician however notices that the patient was born with a horseshoe kidney, which lies just inferior to the vessel at the tip of the yellow arrow. What is this structure?



A. Superior Mesenteric Artery
**B. Inferior Mesenteric Artery**
C. Gastroduodenal Artery
D. Superior Mesenteric Vein
E. Left Renal Vein

**Explanation:** The yellow arrow is pointing to the inferior mesenteric artery. Veins would be excluded because one can observe the presence of atherosclerotic plaques (in white) in the vessel, which would suggest an artery. Horseshoe kidneys are positioned just below the inferior mesenteric artery (this was taught in the CPR module).

22. A 28-year old man has not eaten for 30 hours. A sample of his tissues shows elevated AMP levels. The aforementioned compound is a direct activator of which of the following enzymes during this period?
A. Glycogen Synthase
B. Hepatic Glycogen Phosphorylase
C. Hepatic Glycogen Phosphorylase Kinase
D. Muscle Glycogen Phosphorylase Kinase
E. **Muscle Glycogen Phosphorylase**

**Explanation:** After a 30-hour fast, the man would have significant amounts of glycogenolysis. AMP is a direct activator of glycogenolysis in skeletal muscle only, and acts on the enzyme glycogen phosphorylase.

23. A 43-year old woman is admitted to the hospital with upper abdominal pain. A gastroscopic examination reveals multiple small ulcerations in the body of the stomach. Which of the following nerves transmits the sensation of pain from this region?
**A. Greater thoracic splanchnic nerves**
B. Lumbar splanchnic nerves
C. Spinal nerves T12-L2
D. Vagus Nerve
E. Lesser thoracic splanchnic nerves

**Explanation:** The stomach is derived from the foregut, which is supplied by greater thoracic splanchnic nerves (sympathetic) and the vagus nerve (parasympathetic). Proximal to the midpoint of the sigmoid colon, pain fibers travel with sympathetics.

24. Genetic testing is done for an entire family. It is determined that their child, who is 8-years old, has a mutation in his *HFE* gene. Which of the following will most likely be seen as he grows older?
A. Keyser-Fleisher rings
B. Hair will become twisted, grayish and "kinky"
C. Decreased ceruloplasmin in his serum
**D. Excessive accumulation of iron in his liver and pancreas**
E. Reddish skin pigmentation (this is also correct)

**Explanation:** A mutation of the HFE results in hemochromatosis, which causes accumulation of iron in the liver and pancreas.

25. A 17-year-old adolescent boy who is being treated with the macrolide antibiotic erythromycin complains of nausea, intestinal cramping, and diarrhea. The side effects are the result of the antibiotic binding to receptors in the GI tract that recognize which gastrointestinal hormone?
A. Cholecystokinin
B. Enterogastrone
C. Secretin
**D. Motilin**
E. Gastrin

26. A 63-year old man with a history of alcoholism is brought to the emergency department with hematemesis. Findings on endoscopic examination suggest bleeding from esophageal varices. The varices are most likely a result of the anastomoses between the left gastric vein and which other vessel or vessels?
**A. Azygos system of veins**
B. Subcostal veins
C. Left umbilical vein
D. Inferior vena cava
E. Superior mesenteric vein

27. A 12-year old boy presents with lethargy, weakness, gout and hepatomegaly. Bloodwork shows elevated levels of lactate and a fasting blood glucose level of 40 mg/dL (normal: 70-110 mg/dL). Which one of the following enzymes is most likely to be deficient in this boy?
A. Myophosphorylase
B. Acid Maltase
C. Phosphoglucomutase
D. 4:6 transferase
**E. Glucose-6-phosphatase**

**Explanation:** The boy presents with weakness, gout (increased uric acid), hepatomegaly, elevated levels of lactate and is severely hypoglycemic. This is a classic case of Von Gierke disease, in which the deficient enzyme is Glucose-6-phosphatase.

28. A researcher is investigating the effects of a drug that binds to the enzyme adenine nucleotide translocase. This particular drug binds to this enzyme on the intermembrane space side. Which of the following is the drug the researcher is investigating?
A. Bongkrekic acid
B. Aspirin
**C. Atractyloside**
D. Gramicidin
E. 2,4-dinitrophenol

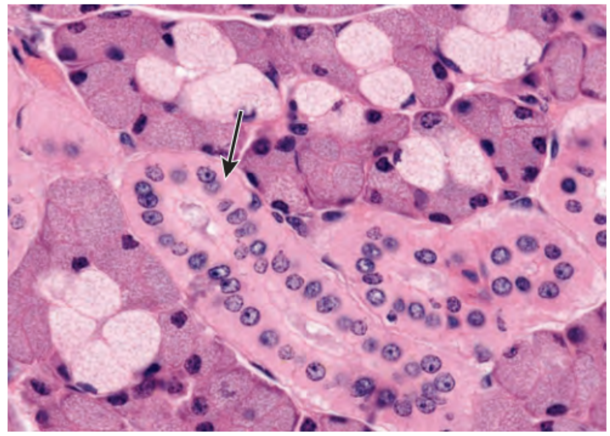
**Explanation:** The intermembrane space side allows the influx of ADP. The drug that prevents this action is atractyloside. Bongkrekic acid acts on the matrix side.

29. A 48-year old man with a 20-year history of alcoholism presents with liver cirrhosis. He presents with asterixis (tremors), slurring of speech, sleepiness, vomiting, cerebral edema and blurring of vision. Bloodwork reveals elevated ammonia levels. Which of the following choices would be the best treatment option for this patient?
**A. Lactulose**
B. Asparaginase Replacement Therapy
C. A high-protein diet
D. Vitamin B6
E. Glutaminase Replacement Therapy

**Explanation:** The patient has hyperammonemia due to liver disease, also known as acquired hyperammonemia. Patients with acquired hyperammonemia are typically treated with a low-protein/high-carbohydrate diet, lactulose, or an antibiotic like neomycin. Lactulose is a digestion-resistance sugar than allows for more nitrogen to be excreted in the feces.

30. A biopsy of a salivary gland is taken from a 35-year old woman. The histological slide shows a segment of the biopsy. Which of the following is the most accurate description of the function?

A. Secretion of $Na^+$
B. Secretion of $Na^+$ and $Cl^-$
C. Reabsorption of $Na^+$ and $K^+$
D. Reabsorption of $K^+$ and $Cl^-$
**E. Secretion of $K^+$ and $HCO_3^-$**



**Explanation:** The image displays a striated duct, which secretes $K^+$ and $HCO_3^-$.

31. A 19-year old man gets stabbed at a lower nightclub. His jejunum is severely damaged and the surgeon decides to cut this segment of his small bowel out. The duodenum and ileum are then reattached together. Due to this injury, the absorption of which of the following will most likely be affected?
A. Iron
B. Bile Salts
C. Vitamin B12
**D. Folate**

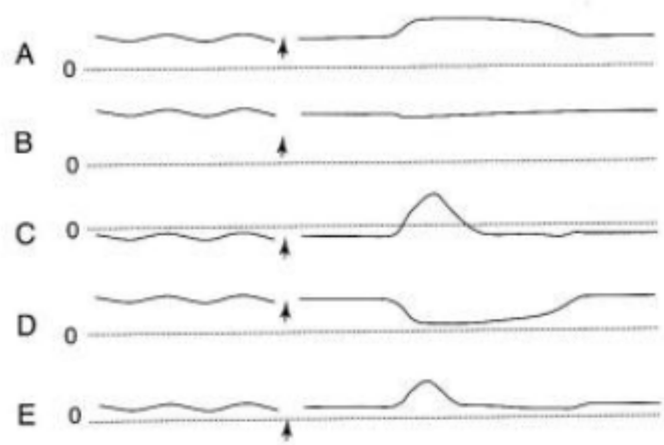
**Explanation:** Iron is absorbed in the duodenum, bile salts and vitamin B12 are absorbed in the ileum. Folate is absorbed in the jejunum (**I**ron **F**ist **B**ro)

32. Which of the following would lead to the most marked increase in the synthesis of primary bile acids by hepatocytes?
A. Increased intestinal bacteria
B. Stimulation by sympathetic nerves
C. Decreased dietary cholesterol
D. An injection of cholecystokinin
**E. Surgical removal of the ileum**

**Explanation:** Bile salts are reabsorbed in the ileum. If the ileum is resected there will be decreased (or absent?) bile salt reabsorption and the liver would have to compensate for this loss by increasing synthesis of bile acids.

33. A 33-year-old man comes to the physician because his chest hurts when he eats, especially when he eats meat. He also belches excessively and has heartburn. His wife complains about bad breath. X-ray after barium swallow shows a dilated esophagus with narrowed LES. Which one of the pressure tracings was most likely taken at the esophagus at mid-thoracic level in this patient before and after swallowing (indicated by arrow)? The dashed line represents a pressure of 0 mm Hg.

A. A
B. B
**C. C**
D. D
E. E




**Explanation:** The mid-thoracic level begins at subatmospheric pressure. The spike later on is likely due to buildup of foodstuffs due to the patients' achalasia.

34. A 4-day old infant boy presents with abdominal distension, constipation and vomiting. An x-ray is taken and can be seen. Which of the following is the embryological cause of the infant's condition?


A. Failure of recanalization of the colon
B. Oligohydramnios
**C. Failure of neural crest cells to migrate into the walls of the colon**
D. Defective rotation of the hindgut
E. Incomplete separation of the cloaca

35. A 44-year old woman presents with fever, night sweats, weight loss, cough and hemoptysis (coughing up blood). She is diagnosed with tuberculosis and is treated with the antibiotic isoniazid. One of the side effects of the aforementioned drug is inhibition of pyridoxal phosphate. As a result, which one of these reactions in heme synthesis would most likely be affected?

A. Protoporphyrin IX → Zinc
B. Hydroxymethylbilane → Uroporphyrinogen III
C. ALA → Porphobilinogen
**D. Succinyl CoA + Glycine → ALA**

E. Porphobilinogen → Hydroxymethylbilane

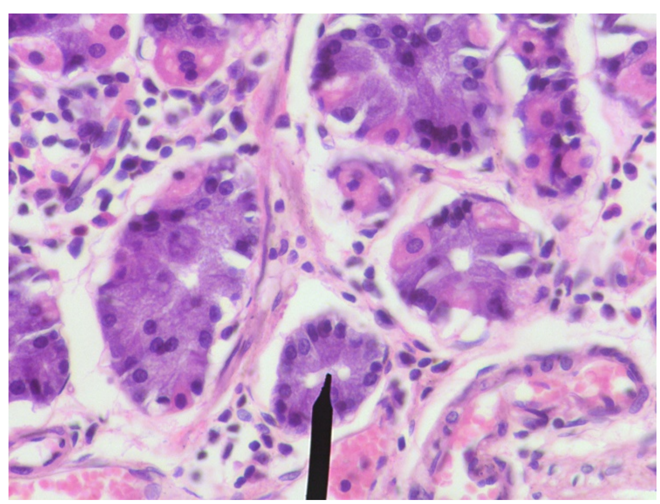
**Explanation:** Pyridoxal phosphate is also known as Vitamin B6, which is a coenzyme required for the enzyme ALA synthase. ALA synthase catalyzes the reaction succinyl CoA + Glycine → ALA, the first and rate-limiting step of heme synthesis.

36. Physical examination of a 60-year-old woman reveals gait instability and decreased proprioception in her lower extremities. Blood tests show a normal hematocrit level and near normal mean cell volume. Her physician orders additional tests. Which of the following laboratory results supports a diagnosis of cobalamin deficiency?
A. Decreased lactate dehydrogenase
B. Negative result of Schilling's test
C. Elevated WBC count
D. Decreased homocysteine levels
**E. Elevated methylmalonic acid**

37. A 52-year old woman has a waist circumference of 41 inches and a BMI of 38 kg/$m^2$. Her physician prescribes her the drug orlistat. Which one of these is the mechanism of the aforementioned drug?
A. Suppresses appetite
**B. Inhibits gastric and pancreatic lipase**
C. Increases fecal excretion
D. An uncoupler of the electron transport chain
E. Inhibits ATP synthase

**Explanation:** The woman in the stem is obese and her physician is prescribing orlistat, a gastric and pancreatic lipase inhibitor to inhibit digestion of dietary fats to reduce dietary TAG absorption. The other pharmalogical drug mentioned in the obesity lecture is sibutramine, which suppresses appetite. Uncouplers such as 2,4-dinitrophenol have also been used to lose weight but may also kill you by accident, so not recommended.

38. A 32-year old female presents to the physician with severe abdominal pain. A biopsy is taken and the following image is seen. Which of the following is secreted by the cells as indicated by the pointer?
A. Pepsin
B. HCl
C. Intrinsic factor
D. Gastrin
E. Mucous
**F. Pepsinogen**



39. A 42-year old man is diagnosed with a tumor of his APUD cells, resulting in carcinoid syndrome. As a result, which of the following conditions may arise?
**A. Pellagra**
B. Cystinuria
C. Pernicious Anemia
D. Achalasia
E. Dubin-Johnson Syndrome

**Explanation:** Carcinoid syndrome is the overproduction of serotonin, which requires tryptophan. Excess shunting of tryptophan to serotonin decrease the amount shunted to make niacin (Vitamin B3). As a result, pellagra (the four Ds: dermatitis, dementia, diarrhea & death) may arise.

40. A 22-year old female student was brought by her friends to the emergency room on New Years' Day. She was sweating profusely, hyperventilating and scared by irregular heartbeats. She was highly agitated and her breath smelled of alcohol. Her friends told the physician, that she had been drinking heavily during the whole night. She did not take medications and had slept for several hours. She suddenly woke up from unusual fast heartbeats.

Her lab results showed the following:

| | | |
|---|---|---|
| Aspartate aminotransferase (AST) | 30 U/L | (reference range 5-35) |
| Alanine aminotransferase (ALT) | 50 U/L | (reference range 5-56) |
| Alkaline phosphatase (ALP) | 70 U/L | (reference range 20-160) |
| γ-glutamyl transpeptidase (GGT) | 157 U/L | (reference range 8-78) |
| | | |
| Total bilirubin | 1.4 mg/dL | (normal <1.4 mg/dL) |
| Direct bilirubin | normal | |
| | | |
| Albumin | 4.3 g/dL | (reference range 3.5-5.5) |

Which of the following is most likely in the clinical scenario?

A. Liver cirrhosis
**B. Lactic academia**
C. Hypoketonemia during fasting
D. Obstructive jaundice
E. Hepatocellular damage

**Explanation:** Her high intake of ethanol resulted in the inability to use lactate by the liver for gluconeogenesis. At the same time, there is generation of more lactate from pyruvate at high NADH levels in liver cytosol, and more lactate is released into the blood. The other choices can be excluded by her blood data or general knowledge of ethanol metabolism.

41. A 67-year-old man with a history of alcohol abuse presents to the emergency room with severe epigastric pain, hypotension, abdominal distension, and diarrhea with steatorrhea. Serum amylase and lipase are found to be greater than normal, leading to a diagnosis of pancreatitis.

The steatorrhea can be accounted for by a decrease in the intraluminal concentration of which of the following pancreatic enzymes?
A. Trypsin
B. Chymotrypsin
C. Colipase
D. Amylase
**E. Lipase**

**Explanation:** Steatorrhea = fatty stools, resulting from an inability to metabolize fat. This may be caused by a deficiency of lipase.

42. A 27-year-old man enters the emergency department in an agitated state. He complains of severe abdominal pain, but soon becomes paranoid and combative, requiring five-point restraint. His vital signs show elevated blood pressure and tachycardia. When a straight catheter is inserted, dark urine flows into the Foley bag. The urine is negative for RBCs, and a toxicity screen result is negative. The doctor suspects a porphyria. Laboratory tests for urine porphobilinogen are positive. Which of the following enzyme deficiencies is most likely responsible for this patient's disorder?
A. Uroporphyrinogen III cosynthase
**B. Porphobilinogen deaminase**
C. Uroporphyrinogen decarboxylase
D. Heme oxygenase
E. Aminolevulinate synthase
F. Ferrochelatase
G. Aminolevulinate dehydratase

43. A 54-year old man with a history of hypertension and hyperlipidemia is brought to the emergency department with acute onset abdominal pain. The pain began at about 6/10 and has been increasing in severity. He reports that for the past few months he has noted abdominal pain about an hour after eating, but not at other times. As a result, he has limited his food intake and lost about 15 pounds over the last 3 months. However, the current pain is much more severe and he reports not having eaten yet. Physical examination is significant for rebound tenderness in the abdomen. The physician decides to pursue mesenteric angiography, which shows significant infarction of the small bowel after thrombotic occlusion of jejunal branches of the superior mesenteric artery. The study also shows occlusion of the celiac trunk; however, there is no evidence of infarction in organs supplied by that artery. Anastomoses between which of the following pairs of arteries best explains the lack of infarction in these organs?
A. Left gastric artery and right gastric artery
B. Left gastro-omental (gastroepiploic) artery and right gastro-omental (gastroepiploic) artery
C. Proper hepatic artery and gastroduodenal artery
D. Right colic artery and middle colic artery
**E. Superior pancreaticoduodenal artery and inferior pancreaticoduodenal artery**

**Explanation:** The superior pancreaticoduodenal artery is a branch of the gastroduodenal artery, which is a branch of the common hepatic artery, itself a branch of the celiac trunk. The inferior pancreaticoduodenal artery is a branch of the superior mesenteric artery. These two arteries have extensive anastomoses within the head of the pancreas. With occlusion of the celiac trunk, blood from the superior mesenteric artery would reach the branches of the celiac trunk via these anastomoses.

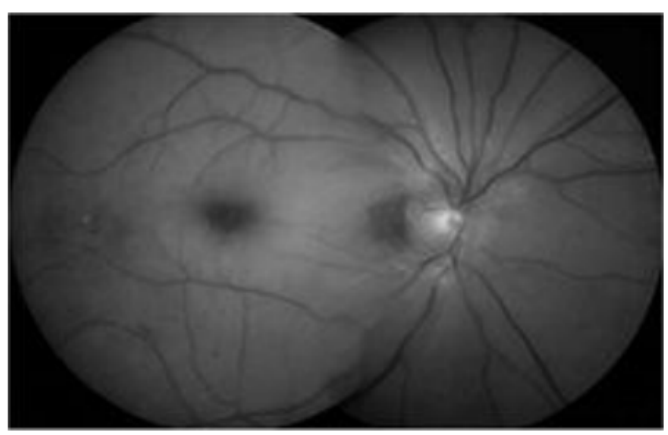
44. A 74-year old female patient with a large gastric carcinoma underwent a total gastrectomy (removal of the stomach). Without supportive therapy, she is likely to develop which of the following?
A. Inability to regulate bile secretion
**B. Pernicious anemia due to Vitamin B12 deficiency**
C. Vitamin D deficiency
D. Increased fat absorption
E. Metabolic alkalosis

**Explanation:** Parietal cells in the stomach secrete intrinsic factor, which is required for the absorption of Vitamin B12.

45. Colipase is a coenzyme to which of the following enzymes below?
A. Salivary amylase
B. Enteropeptidase
**C. Pancreatic lipase**
D. Trypsin
E. Pepsin

**Explanation:** Important to note that colipase has no enzymatic activity in of itself, it just serves as a coenzyme to pancreatic lipase.

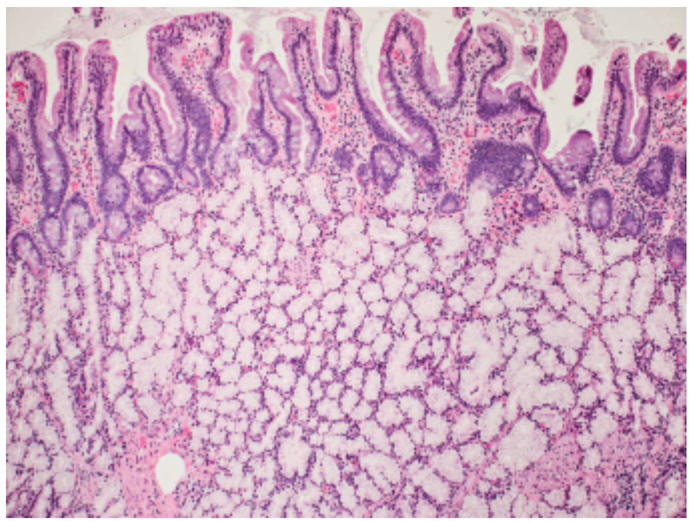
46. The wife in an Ashkenazi Jewish family brings her 1-year-old daughter to the pediatrician. Her previous pregnancy was uneventful and resulted in a full-term healthy girl who is now 4 years old. Her younger daughter, however, has demonstrated a progressive series of behaviors over the first year of life. Her motor skills have diminished and she demonstrates an increased startle reaction. Physical examination is notable only for the ocular findings shown in the image. Deficiency of which enzyme is responsible for this disease?
(A) α-Galactosidase A
(B) Arylsulfatase A
**(C) Hexosaminidase A**
(D) Iduronate sulfatase
(E) Lysyl hydryolase



**Explanation:** Ashkenazi Jews are particularly susceptible to Tay Sachs disease, a lysosomal storage disorder due to deficiency of Hexoasamindase A. A "cherry-red macula" can be seen in the image (well it's a black and white image so you can't see red per say, but the macula is darker than usual).

47. In the small intestine, Na+ is absorbed by each of the following processes *except*
A. diffusion
B. coupled to amino acid absorption
C. coupled to galactose absorption
D. coupled to the transport of H+ in the opposite direction
**E. coupled to the absorption of HCO3-**

48. A 49-year old male comes to the physician complaining of severe abdominal pain, especially after eating a meal. Imaging is done and a biopsy is taken from the affected area. Which of the following locations was the biopsy taken from, based off the histological side?
A. Ileum
B. Stomach
C. Large intestine
D. Jejunum
**E. Duodenum**



**Explanation:** It is the duodenum because you can see the presence of Brunner's Glands.

49. A 74-year-old woman is admitted to the emergency department with severe diarrhea. An arteriogram reveals 85% blockage at the origin of the inferior mesenteric artery from the aorta. Which of the following arteries would most likely provide collateral supply to the descending colon?
A. Splenic artery
B. Sigmoid artery
**C. Middle colic artery**
D. Left gastroepiploic artery
E. Superior rectal artery

50. A 7-day old infant is brought to his pediatrician because of lethargy and feeding problems. He is admitted to the hospital for failure to thrive. Shortly after admission the infant has a seizure. A biochemistry evaluation reveals a deficiency in alpha-ketoacid dehydrogenase. To prevent intellectual disability and death, intake of which of the following amino acids should be restricted in this patient?

A. Alanine
**B. Leucine**
C. Threonine
D. Tryptophan
E. Tyrosine

**Explanation:** The child has a deficiency of alpha-ketoacid dehydrogenase, which results in maple syrup urine disease. As a result, he is unable to metabolize branched chain amino acids. This includes leucine, isoleucine and valine.

51. A 1-year old boy is brought to the physician by his parents because of severe hypotonia. The patient's mother states that he has difficulty feeding because of progressive dyspnea and fatigue. Physical examination shows a liver edge about 5cm below the right costal margin, and rales are heard on auscultation of both lower lung fields. A chest x-ray shows cardiomegaly. Fasting plasma glucose plasma levels are within the normal range. Deficiency of which of the following enzymes is most likely responsible for the patient's symptoms?
A. Glucose-6-phosphatase
B. Glycogen branching enzyme
**C. Lysosomal glucosidase**
D. Lysosomal sphingomyelinase
E. Myophosphorylase

**Explanation:** The boy shows hypotonia, dyspnea, fatigue, hepatomegaly and **cardiomegaly** (cardiomegaly in particular is a strong diagnostic factor). This is characteristic of Pompe disease, which results in the accumulation of glycogen in lysosomes. In Pompe disease, the affected enzyme is Lysosomal glucosidase, also known as acid maltose.

52. A 5-day-old boy is brought to the emergency department after a tonic-clonic seizure at home. The infant is the product of a full-term, uneventful pregnancy, and was normal until two days prior to presentation. The mother reports irritability and poor feeding at home, and the infant was difficult to rouse this morning before suffering the seizure. On physical examination, the infant is tachypneic to 75/min, has icteric sclerae, and has poor muscle tone throughout. Laboratory studies show the following levels: plasma ammonia, 300 pmol/L (normal = 10-40 pmol/L); blood urea nitrogen, 1.5 mg/dL; and creatinine, 0.4 mg/dL. A plasma amino acid analysis fails to detect citrulline. Urine amino acids demonstrate elevated orotic acid levels. This patient suffers from a deficiency of which of the following enzymes?
A. Argininosuccinate lyase
B. Arginase
C. Argininosuccinate synthetase
D. Carbamoyl-phosphate synthetase I
**E. Ornithine transcarbamoylase**

**Explanation:** The stem specifies the increase of orotic acid levels. The defect could therefore be a step in the urea cycle (ornithine transcarbamoylase) or pyrimidine synthesis (carbamoyl-phosphate synthetase I). However, the stem also specifies elevated plasma ammonia, so we know the answer is E.

53. A 12-year old year old male is noted to have neurologic developmental delays. His vision has been getting worse, and has been showing signs of ataxia over the last several months. His serum levels show an abnormally high count of phytanic acid. Which of the following is defective in this patient?
A. ω-oxidation
B. Peroxisomal oxidation of fatty acids
C. MCAD deficiency
**D. α-oxidation**
E. Methylmalonyl CoA mutase

**Explanation:** An abnormally high count of phytanic acid suggests Refsum disease, a disorder of peroxisomes, which is due to a defect in α-oxidation.

54. A new mother calls the pediatrician because she is concerned that her infant defecates after every meal. Which of the following is the cause of these normal bowel movements in newborns?
A. Peristaltic rushes
B. MMCs
C. Gastroileal reflex
D. Rectosphinteric reflex
**E. Gastrocolic reflex**

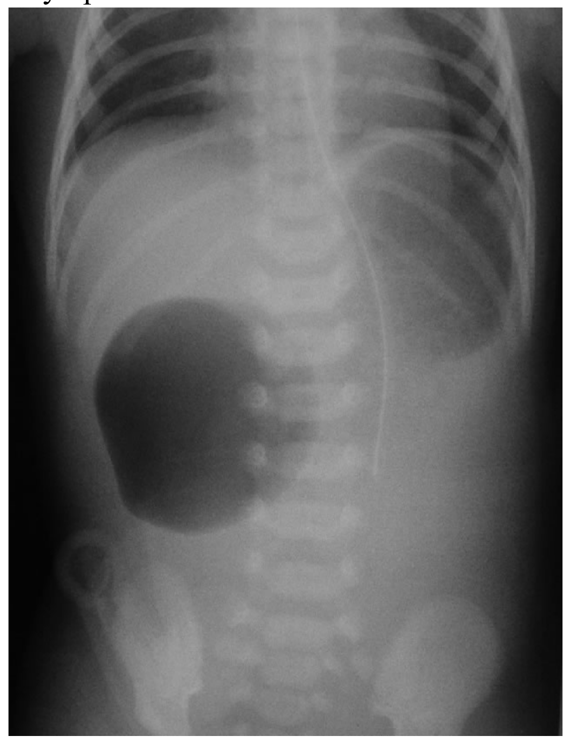
55. A 45-year-old man is found unconscious in an alley by pedestrians who call 911. Emergency medical technicians determine he has a blood glucose level of 35 mg/dL and a blood alcohol concentration of 0.2%. A glucose infusion is started in the ambulance en route to the hospital. After blood glucose levels are stabilized, the patient regains consciousness, but is still intoxicated. He states that he received some distressing news two weeks ago, and drank for two weeks without eating. Laboratory studies show AST, ALT, and serum albumin are all within the normal range. Which of the following most likely contributes to this patient's hypoglycemia?
**A. A high cytoplasmic NADH to NAD+ ratio**
B. A high mitochondrial FADH2 to FAD ratio
C. Increased conversion of glycerol 3-phosphate to dihydroxyacetone phosphate
D. Increased conversion of lactate to pyruvate
E. Increased conversion of malate to oxaloacetate

56. A 32-year old female comes to the physician complaining of "clouded thinking" frequently. In addition, she complains of tingling in her fingers. Upon further questioning, the physician finds out that she has had a drastic change in diet and exercise, trying to live healthier; she has been running 7 miles a day and has been eating vegan for the past several months. Which of the following vitamins is she most likely deficient in?
A. Iron
B. Folate
**C. Cobalamin**
D. Vitamin E
E. B6
F. Riboflavin

Explanation: Vitamin B12 can only be found in animal products. Deficiency results in macrocytic anemia and neurological problems (ex. clouded thinking).

57. A 4-hour old female infant had been diagnosed in utero with polyhydramnios. He has been vomiting stomach contents and bile. Radiographic examinations of the patient are shown. Which of the following conditions will most likely explain the symptoms?

A. Duodenal stenosis
B. Midgut volvulus
**C. Duodenal atresia**
D. Annular pancreas
E. Megacolon



**Explanation:** The image shows a "double bubble" which suggests duodenal atresia.

58. A 56-year old man has a history of heartburn with excessive acid production. Experimental treatments with drugs that block secretagogues of HCl is being considered. Blockade of which of the following secretagogues would reduce HCl secretion?

| | Gastrin | Somatostatin | Acetylcholine | Histamine |
|---|---|---|---|---|
| A | No | No | Yes | Yes |
| B | Yes | No | No | Yes |
| **C** | **Yes** | **No** | **Yes** | **Yes** |
| D | Yes | Yes | Yes | Yes |
| E | Yes | Yes | No | Yes |

59. An investigator is isolating mucin-secreting cells from various tissues to compare surface receptor types on these cells. Which of the following tissues would most likely yield the highest percentage of mucin-secreting cells?

A. Esophageal mucosa
B. Oral mucosa
C. Parotid gland
**D. Sublingual gland**

E. Submandibular gland

60. Vomiting is a complex process that requires coordination of numerous components by the vomiting center located in the medulla. Which of the following occurs during the vomiting act?

| | Lower Esophageal Sphincter | Upper Esophageal Sphincter | Abdominal Muscles | Diaphragm |
|---|---|---|---|---|
| (A) | Contract | Contract | Contract | Contract |
| (B) | Contract | Contract | Relax | Relax |
| (C) | Relax | Contract | Contract | Relax |
| (D) | Relax | Relax | Contract | Contract |
| (E) | Relax | Relax | Relax | Relax |

**Answer is D.**

61. An 18-year old woman decides to get a tattoo for her birthday. Two months later she presents with a fever, RUQ pain, nausea, vomiting and jaundice. Which of the following laboratory values are expected in this patient?
**A. Elevated direct and indirect plasma bilirubin**
B. Elevated direct bilirubin only
C. Elevated indirect bilirubin only
D. Elevated ALP
E. Elevated ALP and GGT

**Explanation:** Infectious hepatitis is a systemic infection predominantly affecting the liver. When jaundice appears, serum bilirubin rises, and, in most instances, total bilirubin is equally divided between the conjugated (direct) and unconjugated (indirect) fractions. The bilirubin in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment. Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or removal of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts. Alkaline phosphatase, which is excreted in bile, increases in patients with jaundice due to bile duct obstruction, but generally normal when the jaundice is due to acute hepatocellular disease. Bile acids are synthesized in the liver by a series of enzymatic steps that also involve cholesterol catabolism. Liver disease decreases bile acid synthesis. ALP elevation alone indicated a bone disease. ALP and GGT together can signify a bile outflow obstruction.

62. Researchers in a laboratory are investigating controlling an enzyme in a biochemical pathway. This enzyme is the first irreversible step in gluconeogenesis. Which of the following is the location, cofactor and allosteric activator of this enzyme?

| | Location | Allosteric activator | Cofactor |
|---|---|---|---|
| A | Mitochondria | Acetyl CoA | Riboflavin |
| B | Endoplasmic reticulum | Acetyl CoA | Biotin |
| C | Mitochondria | Oxaloacetate | Biotin |
| D | Cytosol | Acetyl CoA | Biotin |

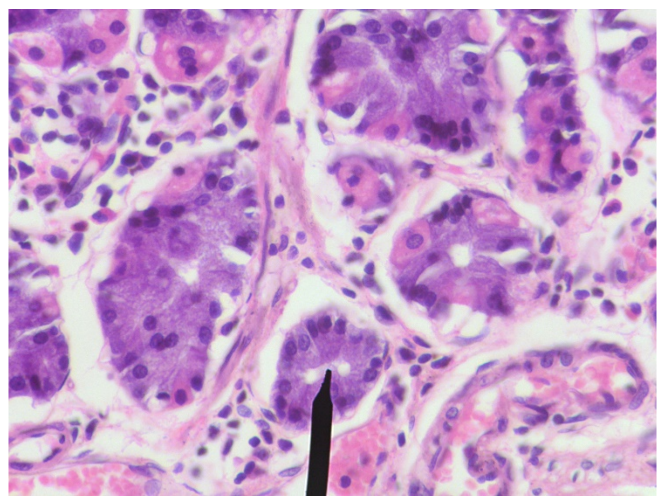
| | | | |
|---|---|---|---|
| E | Cytosol | Acetyl CoA | Thiamine |
| **F** | **Mitochondria** | **Acetyl CoA** | **Biotin** |
| G | Cytosol | Oxaloacetate | Thiamine |

**Explanation:** The first step of gluconeogenesis involves the conversion of pyruvate to oxaloacetate, catalyzed by the enzyme pyruvate carboxylase. This step is located in the mitochondria, requires acetyl CoA as an allosteric activator and biotin as a cofactor (like most carboxylases you will learn in this module).

63. A patient presents to the oncologist for a follow-up visit for small cell carcinoma. The cells in this tumor obtain their ATP mostly from which of the following pathways?
A. Beta-oxidation
B. Electron transport chain
**C. Glycolysis**
D. Gluconeogenesis
E. The Cori Cycle

**Explanation:** The Warburg Effect states that upregulation of HIF by tumor cells induces enzymes of glycolysis, which is the main source of energy for cancer cells.

64. A 32-year old female presents to the physician with severe abdominal pain. A biopsy is taken and the following image is seen. Which of the following is secreted by the cells as indicated by the pointer?
A. Pepsin
B. HCl
C. Intrinsic factor
D. Gastrin
E. Mucous
**F. Pepsinogen**



**Explanation:** The pointer is indicating a chief cell, which secretes pepsinogen. Pepsinogen is later converted to its active form, pepsin.

65. Sanae Ogura, a 25-year old medical student, decides to have a beef burrito from Glover's Grill for lunch. That night, severe diarrhea began, which lasted 3 days. The student produced over 5 liters of watery stools each day and was admitted to hospital on the third day. Her blood pressure was 90/50 mmHg, heart rate was 125 beats per minute and serum potassium was 2.2mEq/l. Which of the following processes was the most likely cause of this student's acute illness?

A. Decreased colonic water absorption.
B. Carbohydrate malabsorption in the small intestine
**C. Increased Cl- secretion by the small intestine**
D. Acute failure of gastric secretion due to gastritis
E. Excess fats in the feces

**Explanation:** The student likely has secretory diarrhea, which is caused by increased Cl- secretion from the crypts due to bacteria or enterotoxin acting on it and increasing the cAMP which is having a positive effect on the CFTR channel and allowing Cl- to flow into the lumen, with Na+ and H2O following passively.

66. A 7-year old boy presents with coarse facial features, airway obstruction, hepatosplenomegaly, developmental delay, gargoylism mildly aggressive behavior and no corneal clouding. Microscopic examination shows the presence of accumulation of GAGs. Which of the following is the most likely diagnosis?
A. I-cell disease
**B. Hunter Syndrome**
C. Fabry Disease
D. Hurler Syndrome
E. Gaucher Disease

**Explanation:** Accumulation of GAGs and the listed symptoms are characteristic of Hunter Syndrome. The key differentiation here from Hurler Syndrome was “no corneal clouding”.

67. A 3-year-old girl is brought to the physician because the mother notices that the child is tripping over objects. Examination of the eyes shows bilateral lens opacification. Urinalysis shows a reducing sugar that is negative for glucose oxidase test. Liver enzymes and serum bilirubin are within normal limits. What is the enzyme responsible for the formation of cataracts in the lens?
A. Sorbitol dehydrogenase
B. Aldolase B
C. Galactokinase
D. Hexokinase
**E. Aldose reductase**
F. Galactose 1-phosphate uridyl transferase

68. Gluconeogenesis begins with the conversion of pyruvate to PEP. Which of the following accurately describes this two-step process?
**A. Oxaloacetate is converted to malate by malate DH within the mitochondria**
B. Acetyl CoA serves as an allosteric activator of PEPCK
C. Pyruvate carboxylase transfers the CO2 on the biotin prosthetic group to pyruvate, which forms oxaloacetate in the cytosol
D. Results in the formation of ATP
E. Binding of glucagon to receptors on liver cells results in the phosphorylation and activation of pyruvate kinase

69. After high intake of ethanol, which of the following steps of gluconeogenesis in the liver is increased due to high cytosolic NADH/NAD+ ratio?
**A. Formation of pyruvate to lactate**
B. Carboxylation of pyruvate
C. Glucose-6-phosphatase
D. Formation of oxaloacetate to malate in the mitochondria
E. Transamination of alanine to pyruvate

70. A 50-year-old malnourished man presents with a 6-month history of night blindness. Physical examination reveals corneal ulceration. The patient is subsequently diagnosed with vitamin A deficiency. Which of the following cells in the liver stores vitamin A as retinyl esters?
A. Kupffer cells
**B. Hepatic stellate cells**
C. Cholangiocytes
D. Endothelial cells
E. Hepatocytes

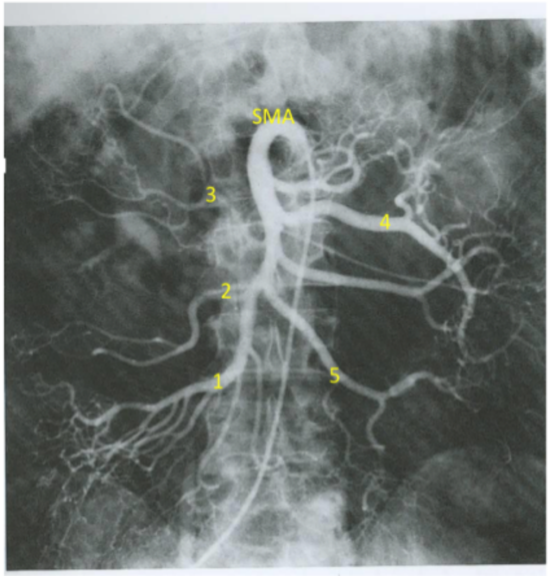
71. A 2-year old child has been admitted for seizures and the following lab reports are obtained. The patient has had previous episodes for the past 6 months. Which of the following enzymes are most likely deficient in this patient?

| | Patient results | Reference ranges |
|---|---|---|
| pH | 7.3 | 7.36-7.42 |
| $HCO_3^-$ | 20 mEq/L | 24-26 mEq/L |
| Fasting blood glucose | 50 mg/dL | 70-90 mg/dL |
| Serum lactate | Increased | |

A. Muscle glycogen phosphorylase
B. Erythrocyte pyruvate kinase
C. PDH complex
**D. Glucose-6-phosphatase**
E. Aldolase B

**Explanation:** The lab results show lactic acidosis and severe fasting hypoglycemia. Additionally, the young child reportedly has seizures. The most likely diagnosis is Von Gierke's disease, which is due to a deficiency of glucose-6-phosphatase.

72. A 27-year old bullfighter is unsuccessful in evading an incoming bull attack and the horns pierce his midsection. The artery indicated #3 is damaged. What is the identity of this artery?



A. Right Colic
**B. Middle Colic**
C. Ileocolic
D. Jejunal
E. Superior rectal

**Explanation:** 1 = ileocolic artery, 2 = right colic artery, 3 = middle colic artery, 4 = jejunal artery, 5 = ileal artery.

73. A 32-year old male undergoes genetic testing out of curiosity and, after reviewing the results with a physician, determines that he has a mutation in a gene encoding an amino acid transporter. However, lab reports show that he has no deficiencies. Which of the following would best explain why?
**A. Peptide fragments are absorbed easily**
B. Amino acids can be absorbed through the stomach
C. Amino acids can be absorbed through colon
D. Intrinsic factor increases uptake of multiple vitamins

**Explanation:** Although the individual amino acid transporter is nonfunctional, peptide fragments can be absorbed and broken down into the amino acid in question.

74. A 15-year old boy appears to his pediatrician for evaluation for polydipsia and polyuria. The patient's blood glucose has been mildly elevated in a chronic state and has abnormally low insulin levels. It is noticed that it requires an abnormally high concentration of glucose in the blood to induce beta cell insulin secretion. Which of the following enzymes is this patient most likely deficient in?
A. Pyruvate kinase
B. Hexokinase
C. Pyruvate dehydrogenase
D. Glucose-6-phosphate dehydrogenase
**E. Glucokinase**

**Explanation:** The patient has chronic mild hypoglycemia and impaired insulin secretion from cells. The diagnosis is MODY-2, or Maturity Onset Diabetes of the Young Type-2, which is a deficiency of the enzyme glucokinase. Patients with this disorder are typically not overweight and require a higher blood glucose level (more than 7-8mM) to trigger insulin release.

75. A 12-year old boy is brought to the physician with severe pain in his abdominal area and recent hematuria. His history shows that he had a complicated birth, but has been seemingly fine since then. The following lab values are seen:

| | Patient | Normal ranges |
|---|---|---|
| Serum ALT | 11 | 8-20 IU/L |
| Serum AST | 12 | 8-20 IU/L |
| Serum ALP | 35 | 20-70 IU/L |
| Serum GGT | 51 | 40-60 IU/L |
| Serum albumin | 4.3 | 3-5 gm/dL |
| Serum bilirubin | .91 | 0.8-1 mg/dL |
| Serum direct bilirubin | .17 | 0.1-0.2 mg/dL |
| Bifringe crystals | Not present | - |
| Hexagonal crystals | Present | - |

Which of the following conditions best explains this patients symptoms?
A. Gilbert syndrome
B. Acute viral hepatitis
**C. Cystinuria**
D. Dubin Johnson syndrome
E. Hemachromatosis
F. Homocystinuria
G. Methylmalonyl CoA mutase deficiency

**Explanation:** The presence of hexagonal crystals is diagnostic for cystinuria, one of the primary causes of kidney stones in children.

76. A 38-year-old woman comes to the physician because her chest hurts when he eats. She also belches excessively and has heartburn. Her husband complains about bad breath. X-ray after barium swallow shows a dilated esophagus with narrowed LES. Which of the following locations usually shows a pressure of subzero in a healthy patient?
A. Upper esophageal sphincter
B. Lower esophageal sphincter
**C. Midline between upper and lower esophageal sphincters**
D. Above the upper esophageal sphincter
E. Below the lower esophageal sphincter

77. A 24-year old woman presents to the ER in a critical state. She had been troubled with a headache earlier in the day and had drank several glasses of red wine and taken an excessively large dose of Tylenol (acetaminophen), which resulted in her current state. What drug would be best suited for this woman?

A. Phenobarbital
B. Disulfiram
C. Fomepizole
**D. N-acetylcysteine**
E. Statin Drugs

**Explanation:** Acetaminophen is metabolized by CYP2E1 (induced by alcohol) to NAPQI. With large doses of acetaminophen and alcohol induction, large amounts of NAPQI may be produced, which bind to SH groups of proteins and can be very toxic, particularly to the liver. The treatment for this is to give N-acetylcysteine, also known as acetadote, which stimulates synthesis of GSH to bind to NADQI. This may prevent severe hepatic damage.

78. The electrical activity of the musculature of the small intestine was monitored aat two positions X and Y. Over a one-minute period, 13 slow waves were recorded at position X and 7 were recorded at position Y. Spike potentials were recorded in 3 of the slow waves at X and 6 of the slow waves at Y. Analysis of these data suggests that:
**A. The number of contractions was less at X than Y**
B. Contractions at X were of longer duration than at Y
C. Segmentation occurred at X and peristalsis occurred at Y
D. The force of contractions was greater at X than Y
E. Position X is located distal to position Y

**Explanation:** Slow waves MAY lead to spike potentials, but it is the spike potentials that are the cause of contractions.

79. A 22-year old woman is involved in a motor vehicle collision. Her complaints are consistent with a pelvic fracture and a CT scan is ordered to rule out internal bleeding. The CT scan shows no bleeding and that her uterus is in the most typical position—anteflexed and anteverted. Anteverted describes the relationship between which of the following structures?
(A) Uterine body and cervix
(B) Uterine body and vagina
**(C) Cervix and vagina**
(D) Vagina and uterine body
(E) Urethra and vagina

80. A new drug Y, directed at blocking gastric H+/K+/ATPase, was tested in healthy volunteers. Which of the following effects would you predict in the presence of drug Y during stimulation of gastric secretion compared to a control subject?
A. Larger volume of gastric secretion
B. Lower K+ concentration in gastric juice
**C. Lower pH of gastric venous blood**
D. A more negative transepithelial voltage
E. Higher H+ concentration in gastric juice

**Explanation:** If the parietal cell isn't secreting H+ into the stomach, then no HCO3- is being produced by carbonic anhydrase and less HCO3- is being secreted into the bloodstream in exchange for Cl- Essentially, no alkaline tide will occur, so the blood will become more acidic.

81. A 31-year old female prize fighter is in critical condition following his previous fight. Due to the inattentiveness of the referee, she received multiple unanswered blows to the head after being knocked out unconscious. After a week in the hospital, she is shown to have increased cardiac output, oxygen consumption, body temperature, energy expenditure, and protein catabolism. Which one of the following phases is she most likely in during this time?
**A. Flow Phase**
B. Anabolic Phase
C. Ebb Phase
D. Hormonal Phase
E. Luteal Phase

**Explanation:** Increased cardiac output, oxygen consumption, body temperature, energy expenditure and protein catabolism suggests she is in the flow phase. Additionally, the timeline also suggests flow phase, as the flow phase typically lasts up to two weeks following initial injury.

82. A 7-year old boy was brought to the emergency department by his mother after he was playing outside in his backyard when he was stung by a bee. He presents with hives on his arms and difficulty breathing. Which of the following reactions had to occur in order to create the compound causing the boy's symptoms?
A. Nitric oxide synthase
B. Glutamate decarboxylase
C. Monoamine oxidase
**D. Histidine decarboxylase**
E. Catechol O-methyl transferase
F. Monoamine oxidase

**Explanation:** Histidine decarboxylase mediates the conversion of histidine to histamine, which is released during allergic reactions.

83. A 34-year old woman is rushed to the emergency department with severe burns all around her body after her apartment caught fire. Physicians stabilize her and she is administered into the ICU. Which of the following statements is accurate of her situation over the next several days to weeks?
A. She will have a positive nitrogen balance
B. Urinary urea is decreased
C. Normalization of C-reactive protein levels indicate a poor prognosis
**D. Her C-reactive protein serum levels will increase**
E. There will be decreased muscle proteolysis

**Explanation:** C-reactive protein is a positive acute phase protein, which will be increased following serious injury.

84. A 2-year old child is hospitalized for evaluation of poor growth and low muscle tone. The most striking physical finding is unruly, “kinky” hair but the child also has increased joint laxity and thin skin. Which of the following enzymes is deficient in this patient, which is the underlying cause for his pathology?
A. Copper transporting ATPase
**B. Lysyl oxidase**
C. Ceruloplasmin
D. Superoxide dismutase
E. Tyrosinase

**Explanation:** The key diagnostic symptom here is “kinky hair”, which is characteristic of Menkes disease. Menkes disease is an inherited defect in copper absorption, which leads to reduced activity of lysyl oxidase. Lysyl oxidase requires copper as a cofactor.

85. A newborn develops marked jaundice and kernicterus within weeks of birth. Blood tests reveal alanine aminotransferase = 16 U/L, aspartate aminotransferase = 14 U/L, and total bilirubin = 3.8 mg/dL. Urine tests are negative for bilirubin. Treatment attempts with phenobarbital, plasmapheresis, and phototherapy are unsuccessful. The infant’s condition deteriorates over the next two months and he dies. What is the underlying cause of the infant’s death?
A. Elevated antimitochondrial antibodies
**B. Problem with bilirubin conjugation**
C. Problem with bilirubin uptake
D. Problem with excretion of conjugated bilirubin
E. Problem with hepatic copper excretion

**Explanation:** The speed of onset (newborn), diagnosis of kernicterus and negative bilirubin in the urine is suggestive of an increase in unconjugated bilirubin. The child likely has physiological jaundice due to low activity of hepatic UDP-glucoronyl transferase.

86. Glucokinase catalyzes the transfer of the gamma-phosphate of ATP to glucose. What is the $\Delta G^{o\prime}$ of the glucokinase catalyzed reaction if given the following standard free energy of hydrolysis values:

$ATP + H2O \rightarrow ADP + P$ $\quad \Delta G^{o\prime} = -30.5$ kJ/mol
$Glc\text{-}6\text{-}P + H2O \rightarrow Glc + P$ $\quad \Delta G^{o\prime} = -14.0$ kJ/mol

A. +16.5 kJ/mol
B. +44.5 kJ/mol
**C. -16.5 kJ/mol**
D. -44.5 kJ/mol
E. Approximately 2 kJ/mol

**Explanation:** The answer is C because you have to flip reaction #2 to cancel out the H2O. Hence, the calculations are -30.5 kJ/mol + 14.0 kJ/mol = -16.5 kJ/mol.

87. A 54-year old woman arrives at her family physician's office for her annual check-up. The physician notices that she has abnormally high levels of cholesterol, and prescribes statin drugs. Statin drugs inhibit formation of which of the following products?
A. Acetyl-CoA
B. HMG-CoA
C. Malonyl-CoA
**D. Mevalonate**
E. 7-alpha-hydrocholesterol

**Explanation:** Statin drugs inhibit the enzyme HMG-CoA reductase, which forms the product mevalonate. This is the rate-limiting step of cholesterol synthesis.

88. A 2-year old boy presents with rugged, werewolf-like features, photosensitivity, red urine and hypertrichosis. A defect in heme synthesis is suspected. Which of the following is most likely the affected enzyme?
A. ALA Synthase
**B. Uroporphyrinogen III synthase**
C. Porphobilinogen deaminase
D. Uroporphyrinogen decarboxylase
E. Ferrochelatase

**Explanation:** The "werewolf" features, photosensitivity and red urine with hypertrichosis suggests Congenital Erythropoietic Porphyria (CEP). The early onset (2 years of age) further supports this notion. The enzyme that is defective in this disorder is uroporphyrinogen III synthase.

89. Following diagnostic imaging showing a "bird-beak's appearance" of the esophagus, a 24-year old man is diagnosed with achalasia. Relaxation of the LES can typically stimulated by nitric oxide (NO). Which of the following amino acids is a precursor for NO?
A. Leucine
B. Glycine
**C. Arginine**
D. Lysine
E. Tryptophan

**Explanation:** Catalyzed by the enzyme nitric oxide synthase, arginine forms NO and citrulline.

90. A 26-year old woman is diagnosed with hyperammonemia and has a large buildup of citrulline in the blood. Which one of these enzymes in the urea cycle is most likely difficult?
A. Carbamoyl phosphate synthetase
B. Ornithine carbamoyltransferase
**C. Argininosuccinate synthetase**
D. Argininosuccinate lyase
E. Arginase

91. A patient presents with altered mental status. The patient's blood glucose is extremely low. A tumor is observed on CT in the beta cells of the pancreas. Which of the following enzymes is most likely directly activated due to the patient's metabolic status?
A. Fructose-2,6 bisphosphatase
B. Glucose 6 phosphatase
C. Pyruvate carboxylase
D. Hexokinase
**E. Phosphofructokinase-2**

**Explanation:** A tumor in the beta cells of the pancreas would increase insulin levels. High insulin levels will activate PFK-2, one of the enzymatic properties of the bifunctional enzyme. This induces the formation to F-2,6 BP which allosterically activates PFK-1 and induces glycolysis.

92. A 44-year old man undergoes a week-long fast to protest increased taxes from the government. During this time, his energy levels are sustained through gluconeogenesis and ketogenesis. Which of the following are both glucogenic AND ketogenic in nature?
A. Arginine
B. Leucine
C. Lysine
**D. Tyrosine**
E. Cysteine

93. A subject was fed a test meal rich in carbohydrates. Within one hour, a rise in plasma insulin concentration was measured. Which hormone is most likely to have stimulated insulin secretion?
A. Vasoactive intestinal polypeptide
B. Cholecystokinin
**C. Glucagon like peptide-1**
D. Gastrin
E. Secretin

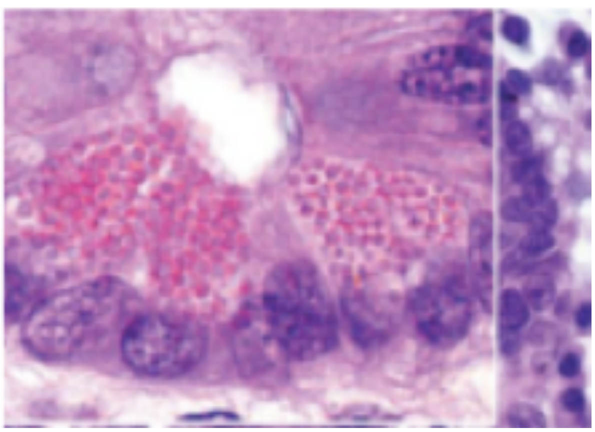
**Explanation:** GLP-1 stimulates insulin release and inhibits glucagon release during ORAL ingestion of glucose.

94. A 28-year old vegan marathon runner presents to the physician with headaches, anemia and methylmalonic aciduria. The accumulation of which of the following amino acids could most likely result in her methylmalonic aciduria?
A. Proline
B. Leucine
C. Lysine
**D. Threonine**
E. Tryptophan

**Explanation:** The patient likely has vitamin B12 deficiency, resulting in her symptoms. In methylmalonic aciduria, there is an inability to metabolize odd-chain amino acids, which include methionine, valine, isoleucine, and threonine.

95. A 6-year old boy lags behind his peers at school in intellectual and emotional development. Further examination reveals a severely enlarged liver and diagnostic tests show the presence of reducing sugar in the urine that is not glucose. What enzyme is most likely deficient in this child?
**A. Galactose 1-phosphate uridyl transferase**
B. Galactokinase
C. Lactase
D. Fructokinase
E. Aldolase B

**Explanation:** The boy shows developmental delay and hepatomegaly, two out of three of classical triad for classical galactosemia, in addition to the presence of a reducing sugar in the urine. One would also expect cataracts, but they do not have to give you all the diagnostics in the stem.

96. The cells on the histological are responsible for producing which of the following substances?

A. HCl and intrinsic factor
**B. Lysozyme**
C. Pepsinogen
D. Motilin
E. Alkaline secretions

**Explanation:** The slide shows Paneth cells, which produce lysozyme (antibacterials) and alpha-defensins (microbicidal peptides)

97. An 18-year old man diagnosed with bulimia nervosa binge-eats an enormous plate of Philly Cheesesteaks. Following this meal, his body releases digestive enzymes to mediate breakdown of the ingested food, which requires relaxation of the Sphincter of Oddi. Which enzymes mediates this relaxation?
A. Gastrin
B. Secretin
C. Somatostatin
**D. Cholecystokinin**
E. Vasoactive Intestinal Polypeptide

**Explanation:** Cholecystokinin (CCK) is stimulated by the presence of fatty acids and amino acids. Its functions include increased pancreatic secretion, increased gallbladder contraction, decreased gastric emptying, and relaxation of the Sphincter of Oddi.

98. A catheter than monitors pressure at its tip is inserted through the nose and passed an unknown distance. The catheter records a pressure between swallows that is subatmospheric and that decreases during inspiration and increases during exhalation. The tip of the catheter is most likely in the:
**A. Esophageal body above the diaphragm**
B. Esophageal body below the diaphragm
C. Upper esophageal sphincter
D. Lower esophageal sphincter
E. Proximal region of the stomach

99. A 34-year old piano mover suffers a serious workplace-related injury and is transported to the hospital. During this time several of his acute phase proteins have increased. Which of the following would NOT be increase?
A. C-reactive protein
**B. Albumin**
C. Haptoglobin
D. Alpha-1 antitrypsin
E. Ceruloplasmin

**Explanation:** C-reactive protein, haptoglobin, alpha-1 antitrypsin and ceruloplasmin are all positive acute phase proteins which would be increased following an inflammatory process during injury. Conversely, albumin is a negative acute phase protein which would decrease.

100. A 42-year old man is pulled over by police following unusual swerving by his car on the road. His blood alcohol levels exceed the allowed limit and in addition, he tries to explain to the police how aliens are chasing him and goes off on a tangential burst of wild stories. He genuinely appears to believe in the aliens and his other wild stories. Which vitamin is likely deficient in this man?
A. Vitamin B6
B. Vitamin B7
C. Vitamin B3
**D. Vitamin B1**
E. Vitamin D

**Explanation:** The patient likely has Wernicke-Korsakoff Syndrome due to excessive alcohol consumption. Symptoms of this disorder include confabulation, where inaccurate, and sometimes wild recollection of events are given, with the patient genuinely believing in the truth of their contents. The vitamin deficient in this syndrome is thiamine, or vitamin B1.

101. A 33-year old woman goes to a small, tropical island for vacation. She ingests unclean drinking water and is subsequently affected by a *V. cholera* infection, resulting in severe secretory diarrhea. Which of the following is characteristic of the aforementioned pathogen?
A. Acts on the CACC channel, increases cAMP
B. Acts on the CACC channel, increases $Ca^{2+}$
**C. Acts on the CFTR channel, increases cAMP**
D. Acts on the CFTR channel, increases $Ca^{2+}$

102. A strain of knockout mice in which the gene encoding phenylethanolamine-N-methyltransferase (PNMT) has been deleted is used to study catecholamine synthesis. The mice are stressed and adrenal glands are dissected and analyzed for catecholamine content. This analysis will most likely show the absence of which of the following in the adrenal glands?
A. Dopa
B. Dopamine
**C. Epinephrine**
D. Norepinephrine
E. Tyrosine

**Explanation:** PNMT catalyzes the conversion of norepinephrine into epinephrine.

103. A 1-year old boy is brought to the physician because of severe hypotonia. The patient's mother states that he has had difficulty feeding because of progressive dyspnea and fatigue. Physical examination shows a liver edge about 5cm below the right costal margin and rates are heard on auscultation of both lower lung fields. A chest x-ray shows cardiomegaly. Fasting plasma glucose levels are within the normal range. Deficiency of which of the following enzymes is most likely responsible for this patient's symptoms?
A. Glucose-6-phosphatase
B. Glycogen branching enzyme
C. Lysosomal sphingomyelinase
**D. Lysosomal alpha-1,4-glucosidase**
E. Myophosphorylase

**Explanation:** The child presents with hepatomegaly, cardiomegaly, and heart failure secondary to cardiomyopathy, all characteristic features of Pompe disease, one of the glycogen storage disorders. This disorder is due to a defect in the enzyme lysosomal alpha-1,4-glucosidase.

104. A 30-year old woman is brought to the emergency department by a friend because of excessive weight loss. For the past three weeks, the woman has participated in a hunger strike as part of a political protest and has consumed only water and vitamin pills. Physical examination shows signs of lethargy and signs of edema. Which of the following will most likely be elevated on laboratory studies of the woman's blood?
**A. Acetoacetic acid**
B. Alanine
C. Bicarbonate
D. Chylomicrons
E. VLDL

**Explanation:** Acetoacetic acid is a ketone, and will be elevated during periods of prolonged fasting.

105. A 7-year old boy is brought to the physician by his mother because of digestive problems. His mother says that he often experiences severe abdominal cramps after eating a high-fat meal. Diagnosis of a genetic defect resulting in a deficiency of lipoprotein lipase is made. Which of the following substances is most likely increased in this patient's plasma following a fatty meal?
A. Albumin-bound free fatty acids
**B. Chylomicrons**
C. HDL
D. LDL
E. Unesterified fatty acids

106. A 7-day old newborn is brought to a physician because of multiple bruises and bleeding at the umbilical stump. The infant has been breast fed and the mother reports taking cephalosporin for a respiratory tract infection she acquired after delivery. Laboratory studies show a prothrombin time that is four-times the reference value. Which of the following biochemical processes is most likely abnormal in this newborn.
A. Conversion of homocysteine to methionine
B. Conversion of methylmalonyl-CoA to succinyl-CoA
C. Degradation of cystathionine
**D. Formation of gamma-carboxyglutamate residues**
E. Hydroxylation of proline

**Explanation:** Deficiency of vitamin K produces a clotting disorder characterized by an elevated prothrombin time and easy bleeding, particularly in neonates. The biochemical basis for this hemorrhagic tendency is that glutamate residues on factors II, VII, IX, and X must be converted to gamma-carboxyglutamate resides for optimal activity.

107. A nondiabetic man consumes a carbohydrate-rich meal. His fasting blood glucose is 80 mg/dL, and his postprandial blood glucose rises to 160 mg/dL. What enzymatic characteristic enables this patient to maintain normal postprandial glucose levels despite ingesting a high carbohydrate load?
A. The high $K_m$ of hexokinase
B**. The high $V_{max}$ of glucokinase**
C. The high $V_{max}$ of hexokinase
D. The long half-life of glucokinase
E. The long half-life of hexokinase
F. The low $K_m$ of glucokinase

108. A 28-year-old wrestler comes to the clinic for a mandated, precompetition physical and blood test. The patient has no medical concerns. Her temperature is 37.0°C (98.6°F), blood pressure is 126/84 mmHg, heart rate is 80/min, and respiratory rate is 14/min. The physical examination is unremarkable. The patient asks about limiting carbohydrate intake to maintain status in her weight class. Which of the following has a role in the regulation of the rate-limiting

step of carbohydrate metabolism?
A. Acetyl-coenzyme A
B. Alanine
**C. Citrate**
D. Glucose-6-phosphate
E. Reduced nicotinamide adenine dinucleotide

109. A 6-month old infant has been exclusively fed a commercially available infant formula. Upon introduction of fruit juices, the child develops jaundice, vomiting, lethargy, irritability, and seizures. Tests for urine-reducing substances are positive. Which of the following sugars should be restricted from the baby's diet?
A. Glucose
B. Galactose
C. Lactose
D. Mannose
**E. Sucrose**

110. Thermogenin is a highly abundant uncoupler in infants, as it plays an important role in thermoregulation by dissipating the H+ gradient in brown adipose mitochondria. Which of the following is most likely to be observed from a mitochondrion in the presence of an uncoupler?
A. Increased H+ concentration in the intermembrane space
B. Decreased ETC activity and decreased $O_2$ consumption
C. Increased ATP synthesis
**D. Increased ETC activity and increased $O_2$ consumption**
E. Decreased $O_2$ consumption only

111. A mother brings her 10-year old daughter to the ER and complains that she keeps trying to eat dirt outside all the time. It was noted that she had pallor and brittle nails and seemed very weak. What could be wrong?
A. High serum ferritin levels
B. High transferrin levels
**C. Low serum ferritin levels**
D. High ceruloplasmin
E. Wilson's Disease

112. Beta-oxidation of fatty acids normally occurs in conjunction with gluconeogenesis and results in the production of NADH and Acetyl-CoA. Acetyl-CoA ensures the functioning of which enzyme?
**A. Pyruvate Carboxylase**
B. PEP Carboxykinase
C. Malate Dehydrogenase
D. Fructose-1,6-bisphosphatase
E. Pyruvate Dehydrogenase

113. Excited to try fresh coconut juice for the first time, a 26-year old medical student runs to the top of the hill one day to purchase some. Unfortunately, the coconut he bought was infested with

*Burkholderia gladioli*, an organism known for producing bongkrekic acid – a respiratory toxin that binds to which enzyme?
A. Cytochrome C Oxidase
B. ATP Synthase
**C. Adenine Nucleotide Translocase**
D. Complex III
E. Cytochrome B

114. A pathologist is examining a liver biopsy specimen taken from a young child. The tissue shows signs of a cirrhotic liver and excess glycogen stores. Analysis of patient's glycogen showed long, unbranched glucose chains. The child most likely has
A. McArdle Syndrome
B. Von Gierke Disease
**C. Andersen Disease**
D. Hers Disease
E. Cori Disease

115. In order to facilitate the digestion, the liver produces bile that allows for emulsification of lipids. What is an example of a secondary bile salt excreted by the liver?
**A. Taurolithocholic Acid**
B. Cholic Acid
C. Deoxycholic Acid
D. Glycocholic Acid
E. Chenodeoxycholic Acid

116. A 42-year old man sees a dermatologist because he has developed vesicles and bullae on his face and arms that appeared after a week-long trip to Florida. His father has a similar condition. A diagnosis of porphyria cutanea tarda is confirmed by finding elevated levels of porphyrins in his serum, urine, and stool. His disease is due to a deficiency of which of the following enzymes?
A. ALA dehydratase
B. Porphobilinogen deaminase
C. Uroporphyrinogen III synthase
D. Ferrochelatase
**E. Uroporphyrinogen decarboxylase**